Proof. testing 1 2 3. □

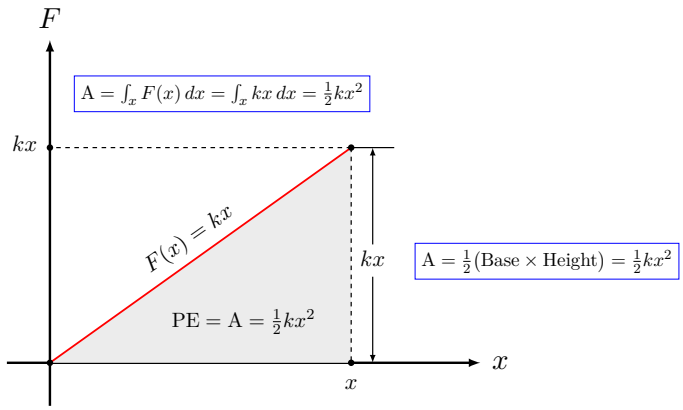

Figure 1: Potential Energy of a Simple Harmonic Oscillator

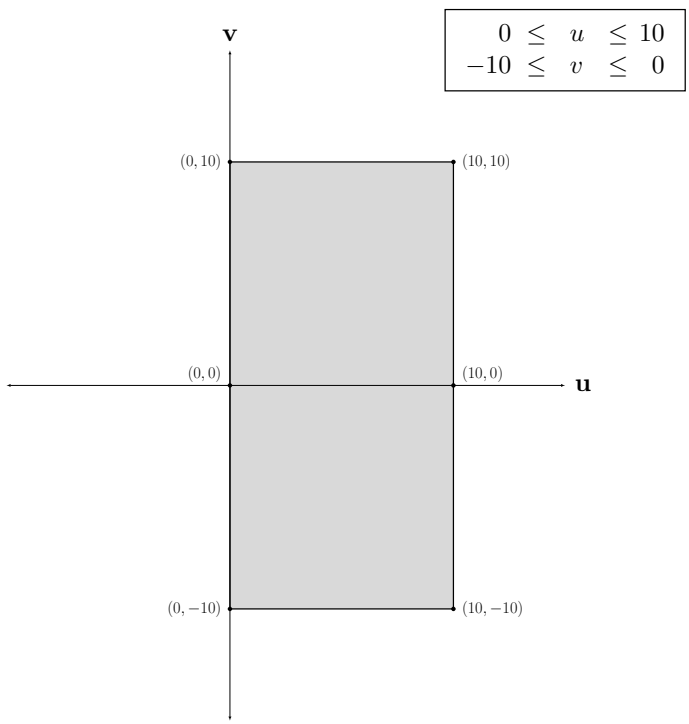

Figure 2: The transformed parallelogram is a rectangle in the uv-plane

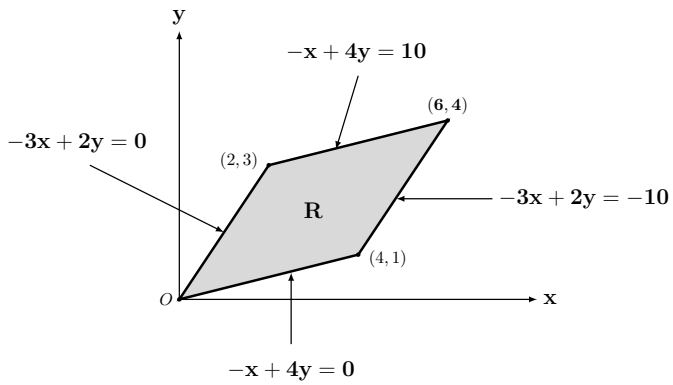

Figure 3: Equations of the Sides of the Parallelogram

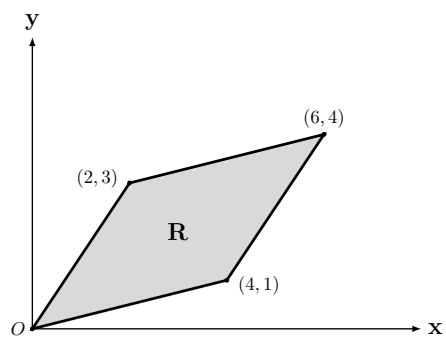

Figure 4: Parallelogram in the xy-plane

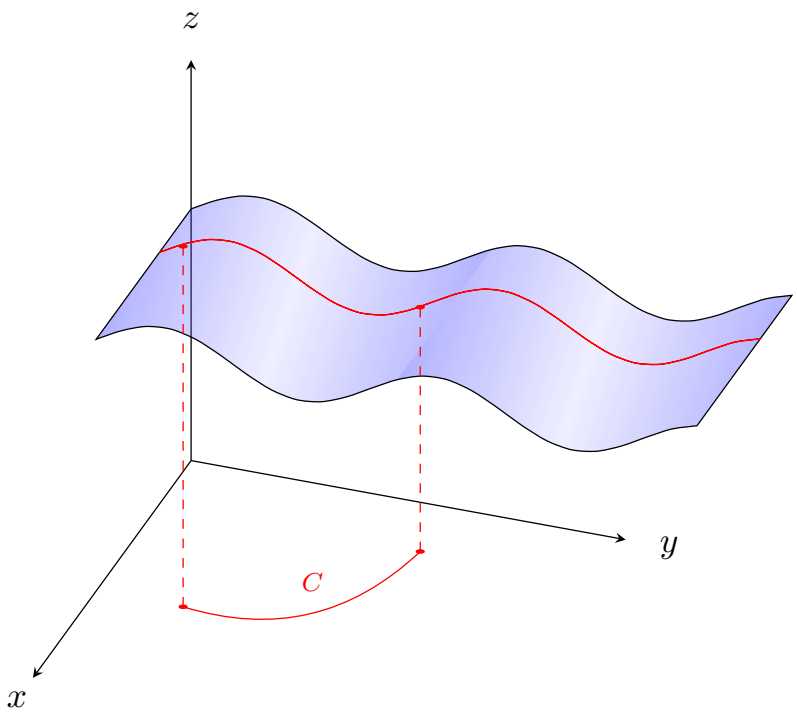

Figure 5: 3D

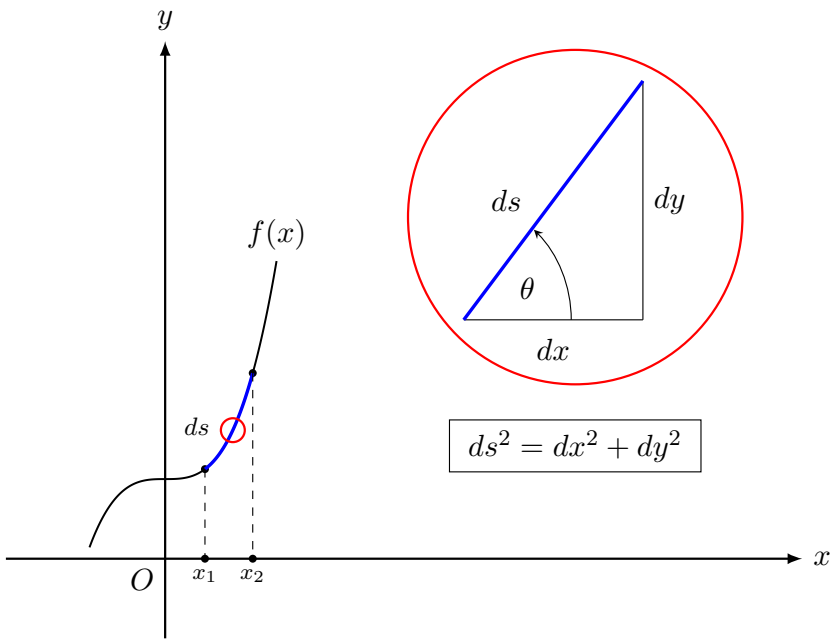

Figure 6: $f(x)$, ds and the Pythagorean Theorem

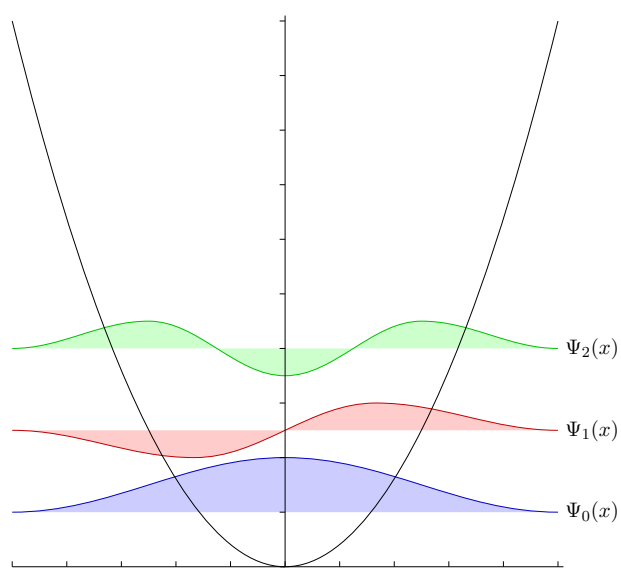

Figure 7: Wave Functions

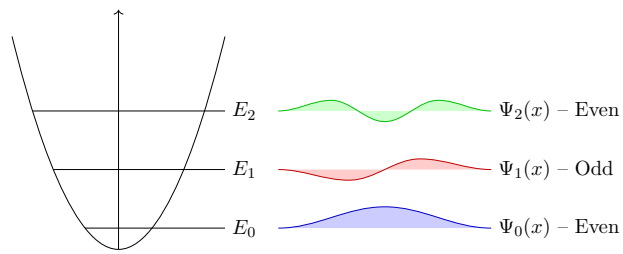

Figure 8: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	81
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 9: A Few Example Kaprekar Numbers

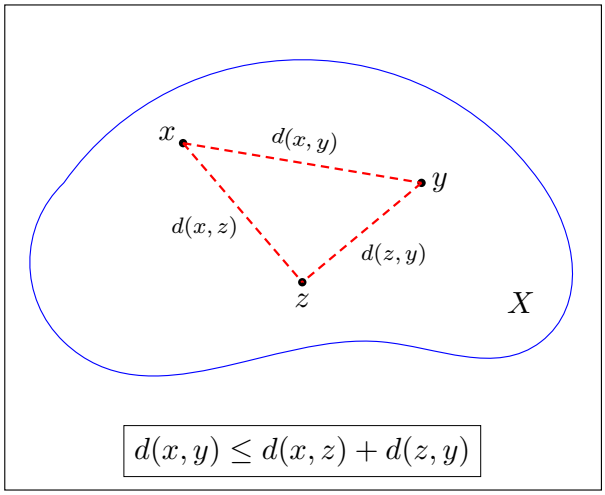

Figure 10: The Triangle Inequality

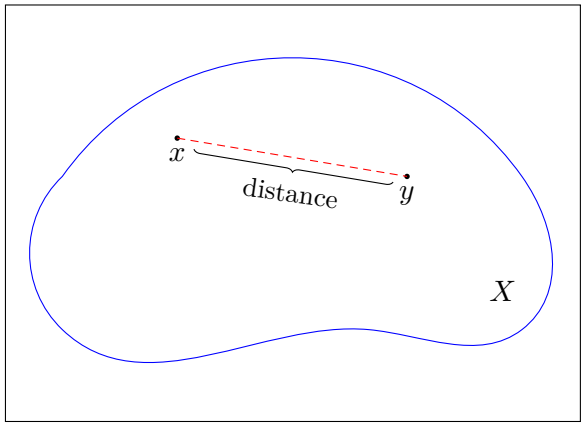

Figure 11: X, x and y

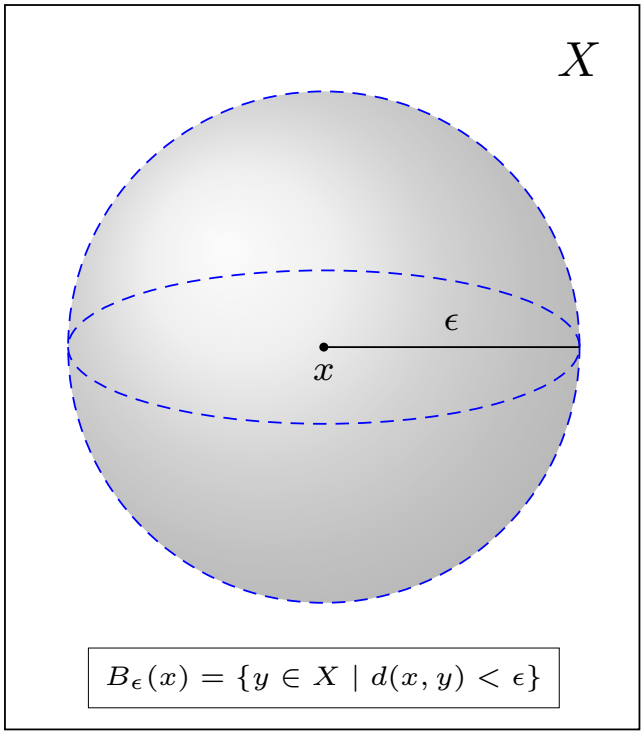

Figure 12: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

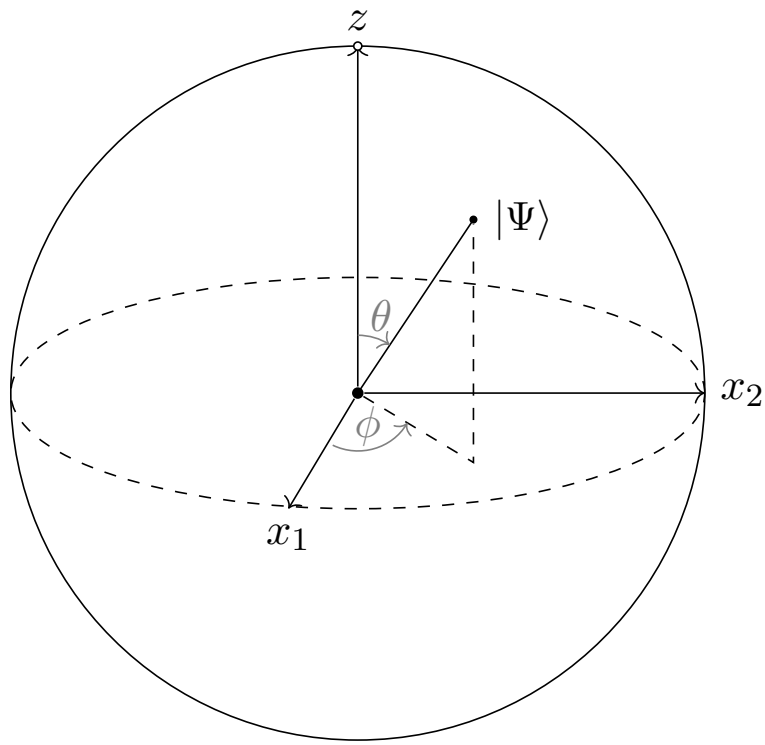

Figure 13: The Bloch Sphere

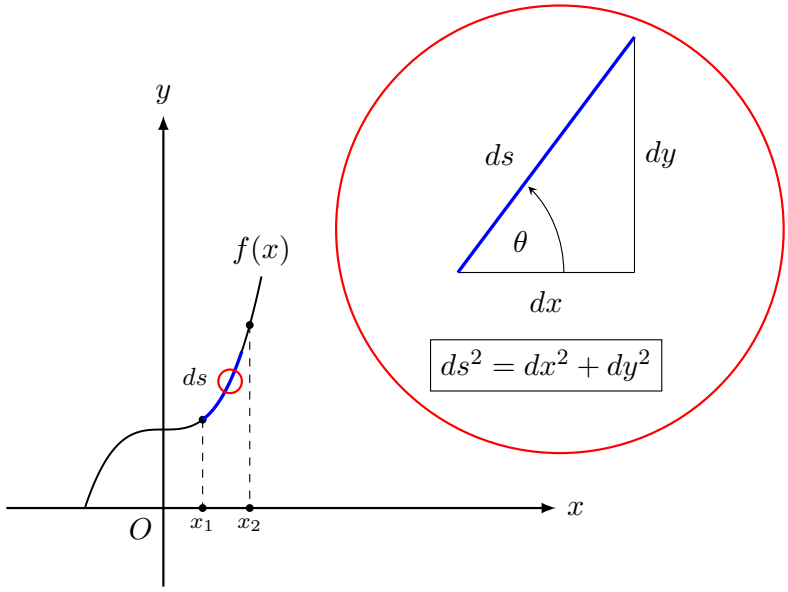

Figure 14: $f(x)$, ds and the Pythagorean Theorem

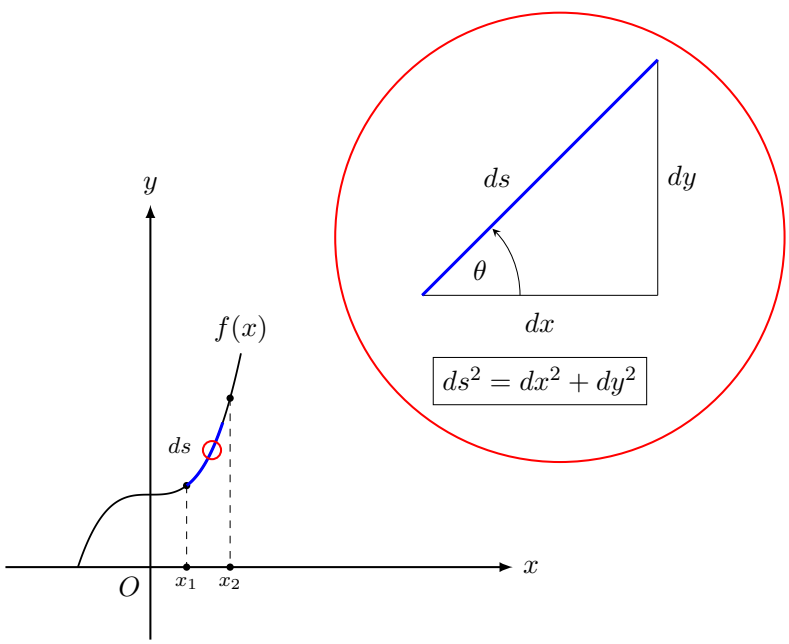

Figure 15: $f(x)$, ds and the Pythagorean Theorem